



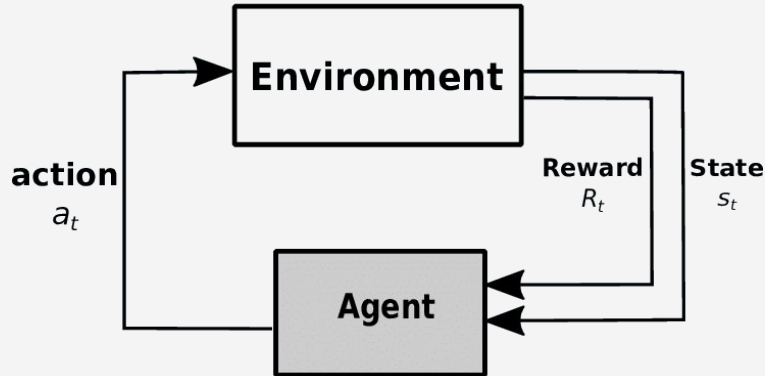
Politecnico
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MARL - Multi Agent RL Introduction

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What is RL?

RL is a machine learning approach where agents learn optimal actions through trial and error, maximizing rewards in an environment. It mimics goal-driven decision-making



From MAS To MARL

MAS: Systems composed of multiple autonomous agents interacting to solve complex problems

+ **Reinforcement Learning**
=

MARL: A subfield of Artificial Intelligence applying reinforcement learning in multi-agent environments



MARL explores **how agents in a MAS learn and optimize behavior** in the presence of other agents

Approaches

Centralized

A central controller makes decisions for all agents based on aggregated information.

Decentralized

Each agent makes independent decisions based on its local information.

Hybrid

Combines centralization and decentralization, where some decisions are made centrally, and others are independent.

Different type of tasks

Cooperative

Agents work together to achieve a common goal.

Competitive

Agents have opposing objectives and compete against each other.

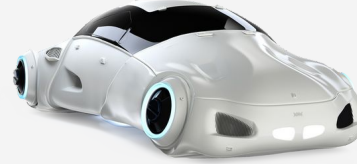
Mixed

A combination of cooperation and competition between agents.

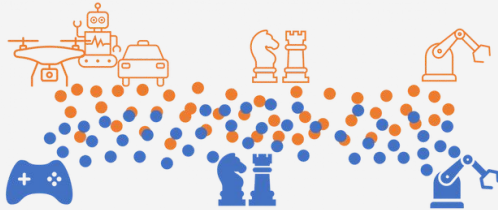
Applications



Robotics and Drone
Coordination



Autonomous Vehicles



Games and
Simulations



Financial Markets



Resource Planning and Management

Case study - Autonomous Warehouse Management

- ✱ **Goal:** To optimize the management of goods through autonomous robots
- ✱ **Agents:** Robots collaborating to move packages
- ✱ **States:** Location of robots, packages, and obstacles
- ✱ **Actions:** Move (forward, backward, right, left), lift/release packages.
- ✱ **Rewards:** Correct delivery of parcels and collisions or delays.

For example, $R(s, a, s') = \begin{cases} +1 & \text{if the package is delivered successfully,} \\ -1 & \text{if collisions or delays occur. (more can be add)} \end{cases}$

Learning Strategies

- ✱ *Cooperation* - > Maximising collective reward
- Decentralized system* -> robots act based their local knowledge
- Communication* -> Exchange of information about the environment



Challenging point



1

Combinatorial Explosion: as the number of agents increases, the state and action space grow exponentially, making learning more complex

2

Non-stationarity: agents are learning while interacting with others, making the environment dynamic and non-stationary which is important for RL alg.

3

Credit Assignment Problem: difficulties in attributing rewards to individual agents when working together, especially in cooperative tasks.

4

Exploration vs. Exploitation: balancing the need for agents to explore new strategies versus exploiting known successful actions.

MARL pros and cons

PROS

1. Information sharing:

- Sensations, episodic experiences or policies learned
- Accelerated learning compared to independent agents

2. Execution of joint tasks:

- Solving complex problems requiring cooperation
- Superior performance in scenarios requiring cooperation

CONS

1. Communication costs:

- Sharing information can be costly.

2. Larger state spaces:

- Collaboration often requires more complex information management

3. Selection of information:

- The relevance of shared information is crucial to avoid interference in learning

Future developments





Thank you!

References

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